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School of Computer Science and Engineering

Working Model

Project Title	AI-Powered Career Counselling and Guidance System for Students: NeoPath
Lab No	AB1-311

Description of Project

Abstract:

This project presents an intelligent career counselling and guidance system designed to assist students in making well-informed career decisions through data-driven insights. The system integrates artificial intelligence, aptitude analysis, and real-time career trend data to provide personalized recommendations. It also offers a secure communication channel allowing students to contact counsellors anonymously through encrypted text, ensuring privacy and emotional comfort. The platform aligns with the objectives of the National Education Policy (NEP) 2020, promoting holistic development and informed career planning.

Key Objectives

• AI-Powered Career Guidance: Use AI algorithms to analyze student profiles, academic records, and aptitude results to recommend suitable career paths.
• Comprehensive Career Database: Provide up-to-date information about various careers, required qualifications, skills, and job market trends.
• Personalized Counselling: Enable one-on-one sessions between students and counsellors to discuss reports, interests, and future goals.
• Anonymous and Encrypted Communication: Introduce a secure text-based counselling feature that uses encryption to maintain confidentiality, allowing students to seek guidance anonymously.
• Interactive Career Assessments: Include aptitude and personality assessments to identify strengths, weaknesses, and potential career fit.
• Career Resource Portal: Host articles, videos, and success stories to help students explore various professional domains.

System Architecture

The system consists of multiple interconnected components designed for secure and intelligent career counselling.

Frontend (Web/Mobile App): Students register, take tests, and interact with counsellors through a user-friendly interface.
Backend (AI Engine): Processes assessment data, matches student profiles with career paths, and generates personalized recommendations using machine learning.
Database: Stores encrypted user data, test results, and session logs using secure cloud infrastructure.
Encryption Module: Protects chat messages and user identity during anonymous counselling sessions using AES/RSA encryption algorithms.
Career Trend Analyzer: Fetches and updates market demand data from verified APIs such as LinkedIn Jobs, Naukri, and government labor databases.
Counsellor Dashboard: Allows authorized counsellors to review student reports, schedule sessions, and communicate securely.

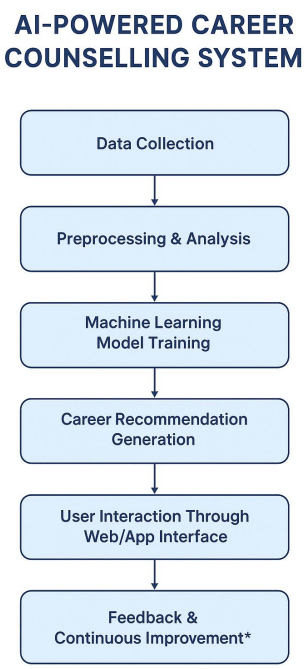
Software Stack

Frontend: HTML, CSS, ReactJS / Flutter (for mobile)
Backend: Python (Flask / Django), Node.js
Database: Firebase / MongoDB
AI Frameworks: TensorFlow, scikit-learn for recommendation models
Encryption Tools: PyCrypto / AES for secure communication
Cloud Services: AWS, Firebase, or Azure for hosting and analytics
Testing & Deployment: Postman, Docker, Jenkins (optional CI/CD pipeline)

Methodology

The project follows a modular, AI-integrated development approach:

1. Student Registration & Profile Creation
2. Assessment Phase
3. Data Analysis & Recommendation
4. Counselling Session (with encrypted chat feature)
5. Report Generation
6. Continuous Learning



Sustainability Focus

• Cloud resources are optimized to reduce energy consumption.
• Lightweight AI models (MobileNet/XGBoost) ensure faster inference.
• Modular upgrades allow integration of new datasets and APIs.
• Anonymous counselling encourages inclusivity and mental well-being.

Alignment with SDGs

SDG 4 – Quality Education
SDG 8 – Decent Work and Economic Growth
SDG 9 – Industry, Innovation, and Infrastructure
SDG 10 – Reduced Inequalities
SDG 16 – Peace, Justice, and Strong Institutions

Testing and Optimization

Functional Testing, AI Validation, Security Testing, and User Experience Testing were conducted to ensure accuracy and usability.

Future Enhancements

• NLP-based voice counselling
• Predictive job placement forecasting

• Multi-language support
• Integration with LinkedIn/Coursera APIs
• Blockchain-based certification

Conclusion

The AI-Powered Career Counselling and Guidance System ensures secure, intelligent, and data-driven career planning, bridging the gap between education and employability.

Faculty and Student Name

Emp. ID / Reg. No.	Name	Role | Department
50926	Dr. Jayanthi R	Associate Professor
23BCE1693	Shriparna Manna	B.Tech Student | Computer Science and Engineering

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